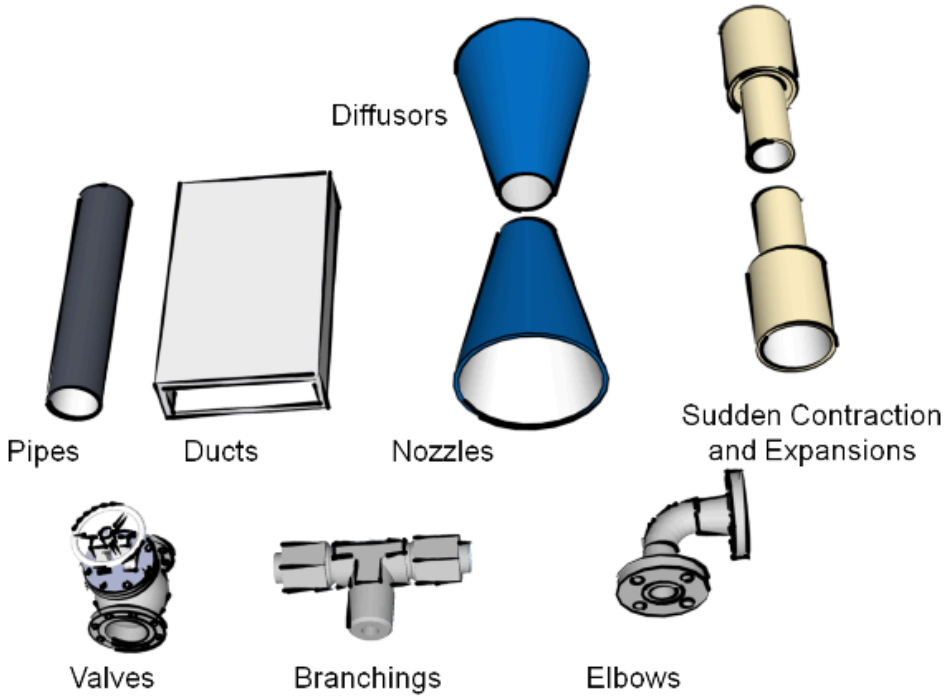
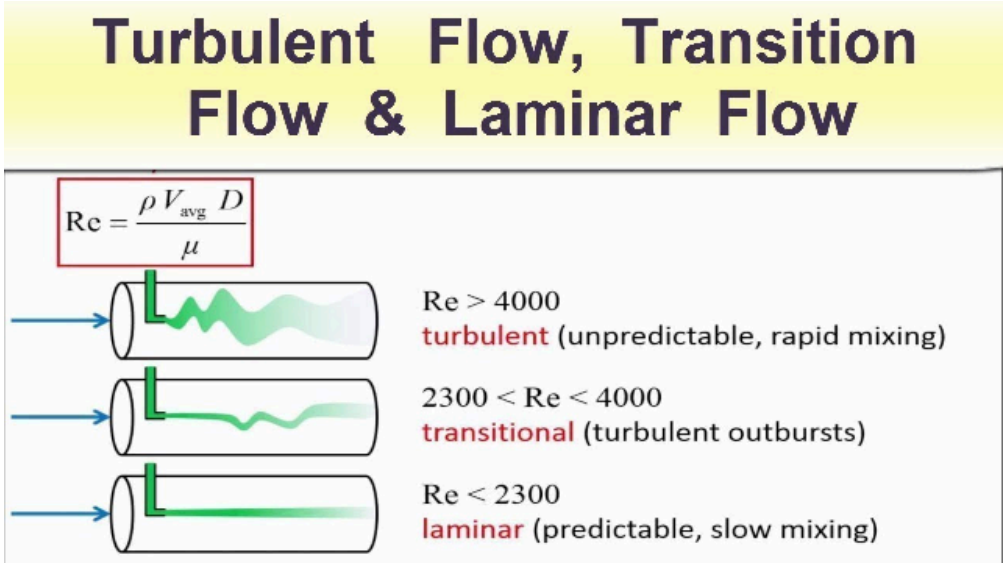
Unit 2
BASIC EQUATIONS OF FLUID FLOW

Characteristics of Internal Flows

- Flows completely bounded by solid surfaces are called **INTERNAL FLOWS** which include flows through **pipes (**Round cross section**),** **ducts (NOT** Round cross section**)**, nozzles, diffusers, sudden contractions and expansions, valves, and fittings.



- The basic principles involved are independent of the cross-sectional shape, although the characteristics of the flow is dependent on it.
- The flow regime (laminar or turbulent) of internal flows (pipe flow) is primarily a function of the Reynolds number (->inertial force/viscous force).
  - Laminar flow: Can be solved analytically.
  - Turbulent flow: Rely heavily on semi-empirical theories and **experimental data.**

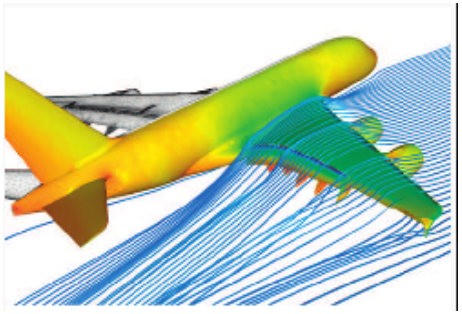


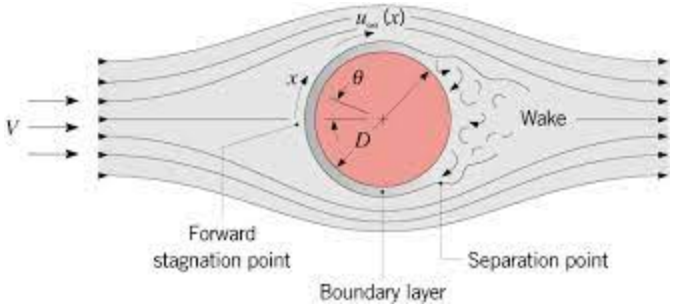
Characteristics of External Flows

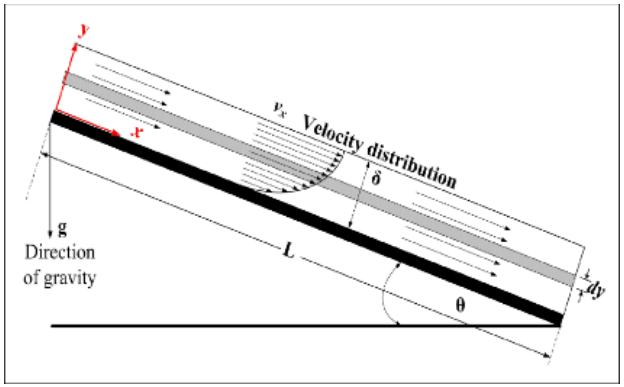
External flow is a flow where boundary layers develop freely, without constraints imposed by adjacent surfaces.

It can be defined as the flow of a fluid around a body that is completely submerged in it.

Examples include fluid motion over a flat plate (inclined or parallel to the free stream velocity) and flow over curved surfaces such as a sphere, cylinder, airfoil, or turbine blade, water flowing around submarines



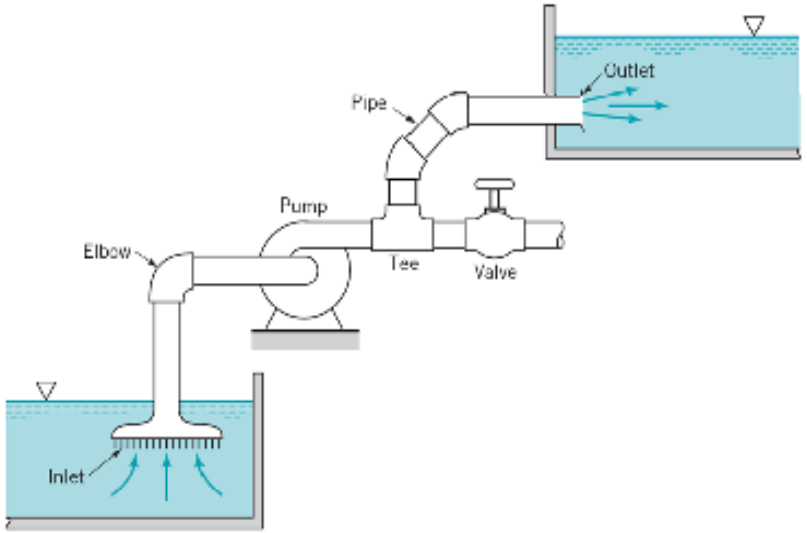






Pipe System

- A pipe system include the pipes themselves (perhaps of more than one diameter), the various fittings, the flowrate control devices valves) , and the pum



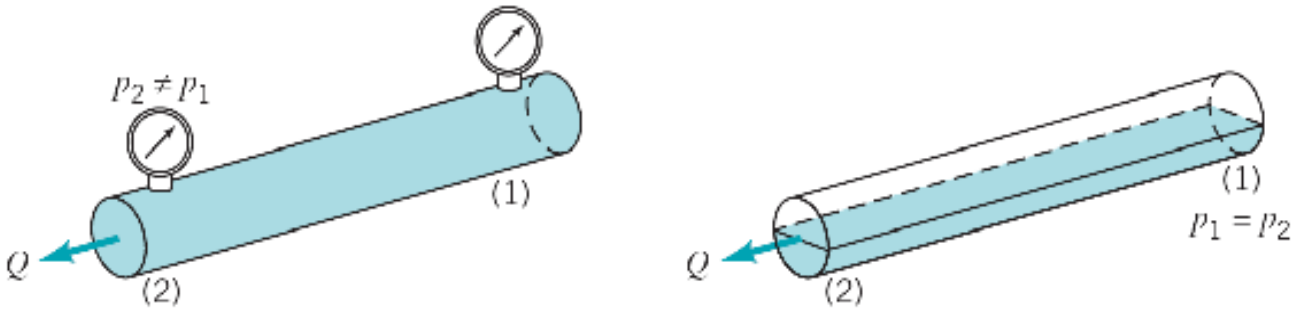
Internal Flow vs. External Channel Flow

- Pipe flow: Flows completely filling the pipe. (a)

  The pressure gradient along the pipe is main driving force.

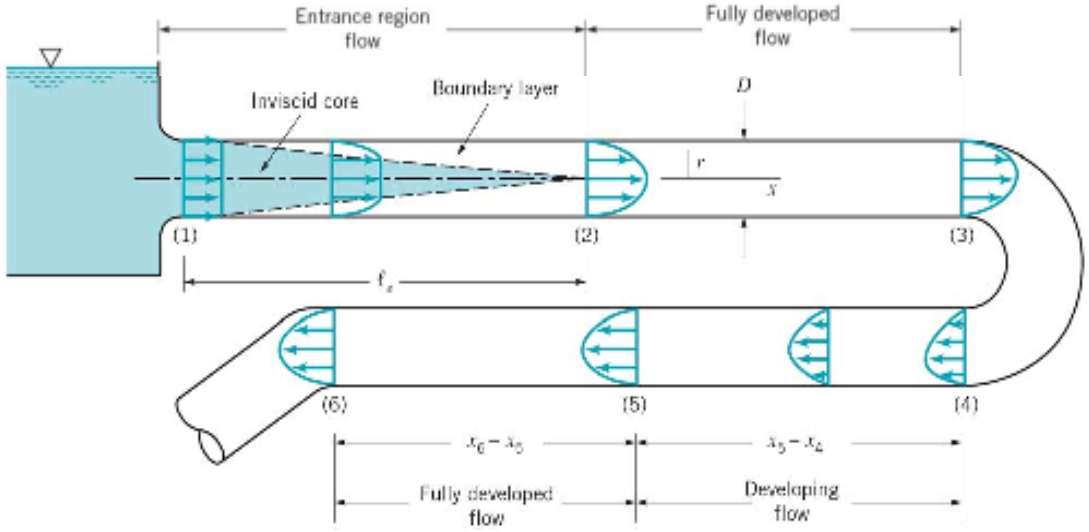
- Open channel flow: Flows without completely filling the pipe. (b)

  The gravity alone is the driving force.



Entrance Region and Fully Developed Flow

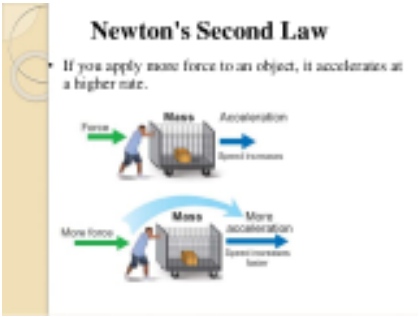
- Any fluid **flowing in a pipe** had to enter the pipe at some location.
- The region of flow near where the fluid enters the pipe is termed the ***entrance (entry) region or developing flow region***..

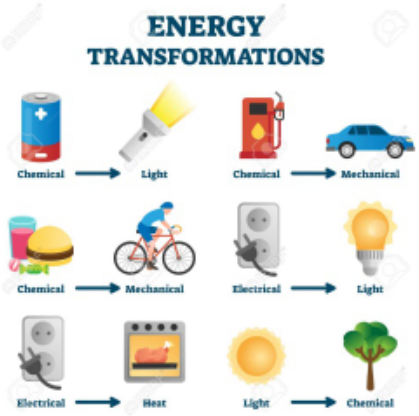


Three basic equations of fluid flow are

- Equation of continuity –Conservation of Mass- Antoine Lavoisier's 1789 discovery that **mass** is neither created nor destroyed in chemical reactions.
- Equation of motion – conservation of momentum - the amount of **momentum** remains constant;
  momentum is neither created nor destroyed, but only changed through the action of forces as described by Newton's laws of motion (2nd law) F=ma
- Bernoulli's equation – conservation of energy - the amount of **energy** remains constant and **energy** is neither created nor destroyed.



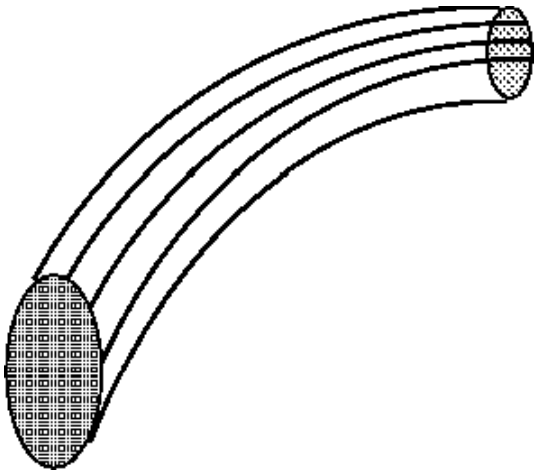




Mass balance in a flowing fluid*:* The rate of mass entering the flowing system equals that leaving as mass can neither be accumulated or depleted with in a flow system under steady conditions.

Continuity equation *:* Consider a stream tube(v is constant) as shown in figure

Let the fluid enter at a point where the area of cross section of the tube is S_a and leaves where the area of cross section is S_b. Let the velocity and density at the entrance be u_a and ρ_a respectively and the corresponding quantities at the exit be u_b and ρ_b.



Assume density in a single cross section is constant. Also assume that the flow through the tube is a potential flow.(no effect due to wall)

Then the velocity u_a is constant across the area S_a and velocity u_b is constant across area S_b.

The mass of fluid entering and leaving the tube per unit time is

$$\dot{m} = \rho_a u_a S_a = \rho_b u_b S_b$$

Where $\dot{m}$ is the rate of flow in mass per unit time.

For a stream tube

$$\dot{m} = \rho\, u\, S = \text{constant}$$

This equation is called the equation of continuity. It applies to both compressible and incompressible flows

For incompressible flow, the above equation reduces to

$$Q = u_a S_a = u_b S_b$$

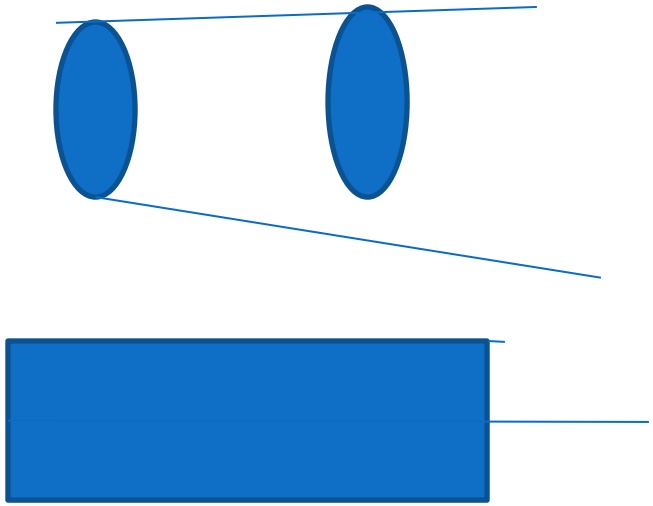
Average velocity : If the flow through the stream tube is not potential flow, the velocity 'u_a' and 'u_b' will vary from point to point across the area S_a and S_b respectively.

Then it is necessary to distinguish between the local and average velocity.

The average velocity of the entire stream flowing through cross section 'S' is defined by V

$$V = \frac{\dot{m}}{\rho S} = \frac{1}{S}\int u\, ds$$

$$V = \frac{\int_0^R \rho u ds}{\rho s}$$



$$V = \frac{\int_0^R \rho u 2\pi r dr}{\rho \pi R^2}$$

also equals the total volumetric flow rate of the fluid divided by the cross sectional area of the conduit. That is

$$V = \frac{Q}{S}$$

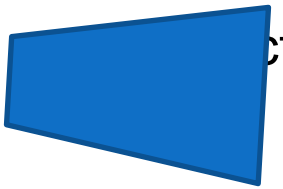
where Q is the volumetric flow rate, m^3/s. ū and u are equal only when the local velocity is the same at all points in area 'S'.

The continuity equation when the velocity varies in a finite stream tube is

$$\dot{m} = \rho_a \bar{u}_a S_a = \rho_b \bar{u}_b S_b = \rho u S$$

Mass Velocity may be written as G, calculated by dividing the mass flow rate by the cross sectional area of channel, [unit kg/sm^2]. The mass velocity 'G' can also be described as the mass current density or mass flux, where flux is defined generally as any quantity passing through an unit area in unit time. The average velocity ū can be described as the volume flux of the fluid.

The diameters of pipe at the section 1 and 2 are 10 cm and 15 cm respectively. Find the discharge through the pipe if the velocity of water flowing through the pipe at section 1 is 5 m/s. Determine the velocity at section 2.

Section 1  ction 2

D_1 = 10cm=0.10 m

D_2 = 15cm=0.15m

$S_1 = (\pi/4)*D_1^2=(\pi/4)\,0.1^2$ = 0.007854m^2

$S_2 = (\pi/4)*D_2^2=(\pi/4)\,0.15^2$ = 0.01767 m^2

v_1= 5 m/s

Discharge through the pipe is given by $Q=S_1v_1$ =0.007854 * 5 =0.03927m^3/s.

continuity equation for incompressible fluid (water)

$v_1S_1\rho_1=v_2S_2\,\rho_2$

$v_1S_1=v_2S_2$

$v_2=(v_1S_1)/S_2$ = (0.00785*5)/0.01767 = **2.22 m/s.**

A 30 cm pipe conveying water, branches into two pipes of diameters 20 cm and 15 cm respectively. If the average velocity in the 30 cm diameter pipe is 2.5 m/s, find the discharge in this pipe. Also determine the velocity in 15 cm pipe if the average velocity in 20 cm pipe is 2m/s.

Data:

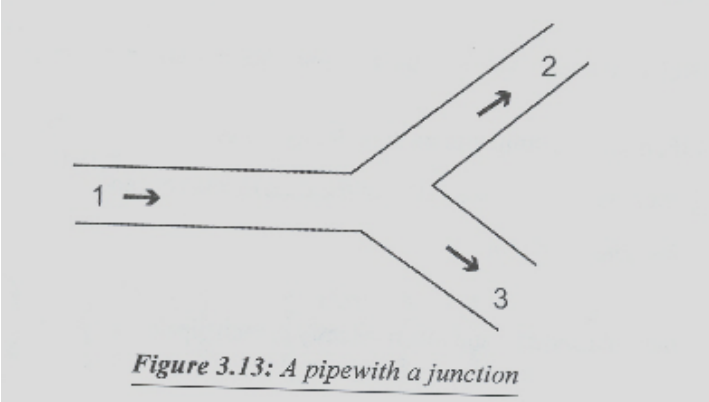


Figure 3.13: A pipewith a junction

$D_1 = 30cm = 0.3m$	$D_2 = 20cm = 0.2m$	$D_3 = 15cm = 0.15m$
$S_1=(\pi/4)*0.3^2= 0.07068\ m^2$	$S_2=(\pi/4)*0.2^2 =0.0314\ m^2$	$S_3=(\pi/4)*0.15^2 = 0.01767\ m^2$
$V_1 = 2.5m/s$	$v_2 = 2m/s$	$v_3 = ?$

$D_1 = 30cm = 0.3m$	$D_2 = 20cm = 0.2m$	$D_3 = 15cm = 0.15m$
$S_1=(\pi/4)*0.3^2= 0.07068\ m^2$	$S_2=(\pi/4)*0.2^2 =0.0314\ m^2$	$S_3=(\pi/4)*0.15^2 = 0.01767\ m^2$
$V_1 = 2.5m/s$	$v_2 = 2m/s$	$v_3 = ?$

Let Q_1,Q_2 and Q_3 are discharges in pipes 1, 2 and 3.
According to continuity equation
$Q_1=Q_2+Q_3$
$Q_1 = S_1v_{1=} = 0.07068*2.5 = 0.1767\ m^3/s$
$Q_2 = S_2v_{2=} = 0.0314*2 = 0.0628\ m^3/s$
\ $Q_3 = Q_1-Q_2 = 0.1767-0.0628=0.1139\ m^3/s$
We know $Q_3 = v_3S_3$
$v_3 = Q_3/S_3 = 0.1139/0.01769 = 6.44$ m/s.

A 25 cm diameter pipe carries oil of specific gravity 0.9 at a velocity of 3m/s. At another section the diameter is 20 cm. Find the velocity at this section and the mass rate of flow of oil.
Data: specific gravity of oil = 0.9 ρ_{oil} = 0.9*1000=900kg/m^3

D_1 = 25cm = 0.25m	D_2 = 20cm = 0.2m
$S_1=(\pi/4)*0.25^2$ =0.049 m^2	$S_2=(\pi/4)*0.2^2=0$ .0314 m^2
v_1 = 3m/s	v_2 = ? ; m =?

According to continuity equation for incompressible fluids at sections 1 and 2

$v_1S_1\rho_1=v_2S\ \rho_2$

$v_1S_1=v_2S_2$

$v_2 = S_1v_1/S_2$ = (0.049*3)/0.0314 = **4.68m/s**

$\dot{m}$ = $\rho_1S_1v_1$ = 900*0.049*3 **=132.3 kg/s.**

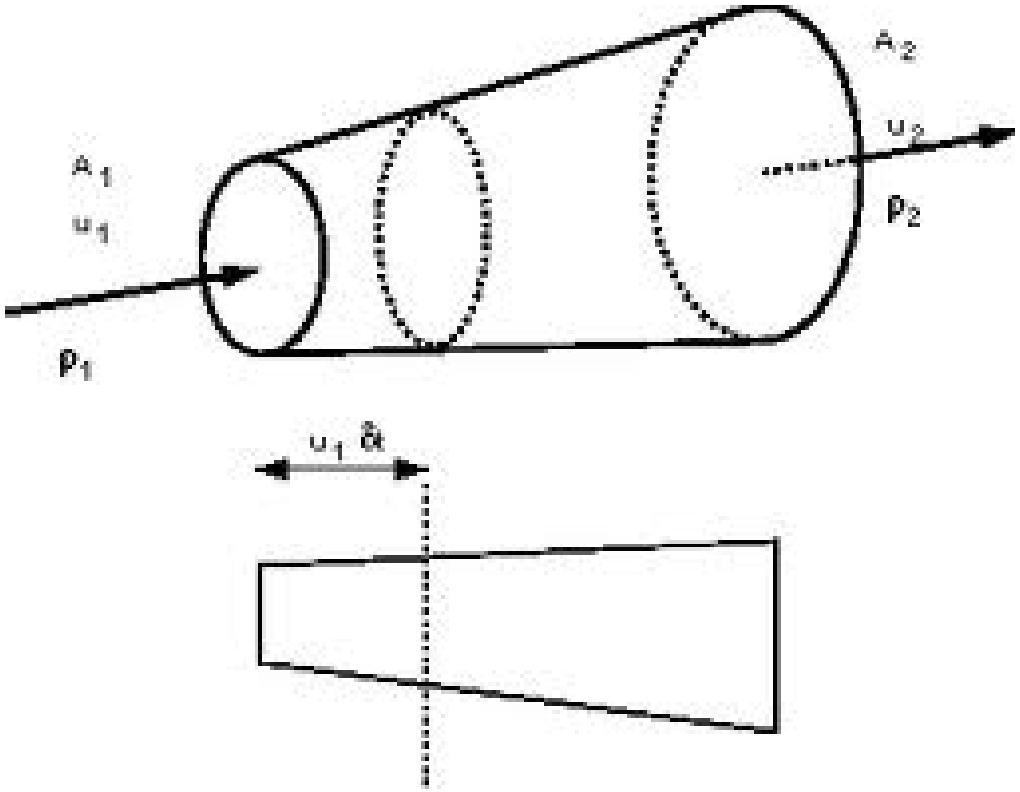
Development of the Momentum Equation

The Rate of change of momentum of a body is equal to the resultant force acting on the body, and takes place in the direction of the force.

To determine the rate of change of momentum for a fluid

consider a streamtube *steady* flow which is *non-uniform*

As shown in figure



In time δt a volume of the fluid moves from the inlet a distance, uδt, the volume entering the streamtube in the time, δt is as follows

Volume entering the stream tube = Area x Distance = $A_1 u_1 \delta t$

Mass entering stream tube = volume x density = $\rho_1 A_1 u_1 \delta t$

Momentum of fluid entering stream tube

= mass x velocity = $\rho_1 A_1 u_1 \delta t u_1$

Similarly, at the exit, we can obtain an expression for the momentum leaving the steam tube:

Momentum of fluid leaving stream tube= $\rho_2 A_2 u_2 \delta t u_2$

We can now calculate the force exerted by the fluid using Newton's 2nd Law. The force is equal to the rate of change of momentum.

Force = rate of change of momentum

$$F = \frac{(\rho_2 A_2 u_2 \delta t u_2 - \rho_1 A_1 u_1 \delta t u_1)}{\delta t}$$

We know from continuity that

$$Q = A_1 u_1 = A_2 u_2$$

and if we have a fluid of constant density

$$\rho_1 = \rho_2 = \rho$$

Then we can write

$$F = Q\rho(u_2 - u_1)$$

Momentum correction factor

Momentum correction factor (β): It is defined as the ratio of momentum of the flow per second based on actual velocity to the momentum of the flow per second based on average velocity across a section.

Why do we use momentum correction factor?

The momentum correction factor (β) is used in fluid flow problems to account for. non-uniform direction of velocities at inlet and outlet sections.

$$\beta =$$

If u = local velocity at distance r. ds the cross sectional area ρ is the density of fluid flowing

Momentum flow per second at r is ρ u ds *u

Total momentum for entire cross section of flow based on actual velocity is (mass flow rate * velocity)

Total momentum for entire cross section of flow based on average velocity is ρ V S * V

For liquids density is constant and hence

$$\beta = \int u^2 dS / V^2 S$$

True momentum per unit time= β x Mass per unit time x Mean velocity

For laminar flow $\beta = \frac{4}{3}$

For Turbulent flow β is slightly above 1 and is upto 1.2

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True momentum per unit time= βx Mass per unit time x Mean velocity

$$\beta = \int u^2 dS / V^2 S$$

For laminar flow $\beta = \frac{4}{3}$

Bernoulli's Equation (or) Theorem

Statement : In a potential flow The total head at any cross section is constant. The total head consists of pressure head, velocity head (kinetic head) and datum head (potential head).

<u>Assumptions</u>

Potential flow: velocity is constant as in a stream tube

Steady flow: flow rate is constant

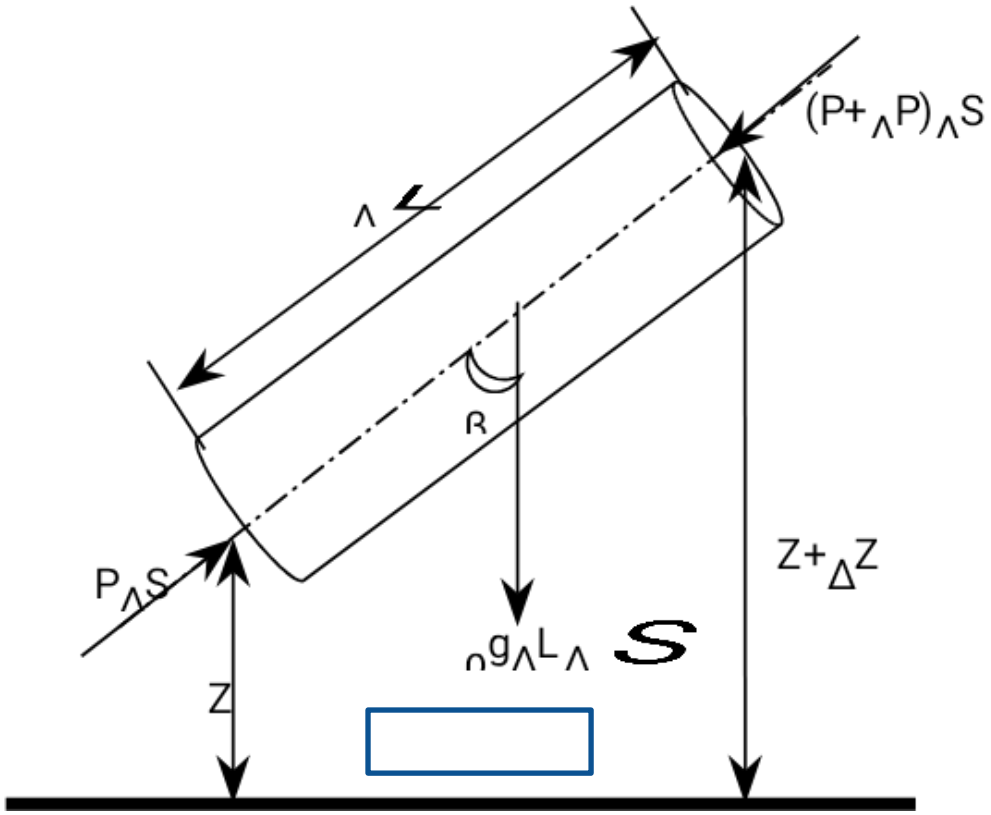
Negligible viscous forces: Ideal fluid

Incompressible fluid : ρ is constant

$$\frac{P}{\rho g} + \frac{u^2}{2g} + z = \text{constan}$$

$$\frac{P_1}{\rho g} + \frac{u_1^{\,2}}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{u_2^{\,2}}{2g} + z_2$$

Consider a volume element of fluid flowing along a stream tube of constant cross section as shown in figure



Let the cross-section of the stream tube be S and average density of the fluid in the element is ρ.

Let P and u are the pressure and velocity at the upstream end of the tube respectively. The pressure and velocity at the downstream end are P + Δ P and u+ Δ u respectively. Axis of the stream tube is making an angle β with vertical. The length of the volume element is Δ L.

The time required for the volume element of fluid to move its own length Δ L is Δ t.

Now consider the forces acting on the element.

The forces, which do act to accelerate or retard flow, are

Pressure Force in the direction of flow = P Δ S

Pressure Force opposing the flow = (P + Δ P) Δ S

The component of force of gravity acting along the axis (in the direction of flow) = ρΔ S Δ L g cosβ

The resultant force in the direction of flow
= PΔS -(P + Δ P) Δ S-ρΔ S Δ L g cos β
This force equals the mass of the volume element multiplied by its acceleration $= \rho \Delta S \Delta L \frac{\Delta u}{\Delta t}$

$$P\Delta S - (P + \Delta P)\Delta S - \rho \Delta S \Delta L g \cos\beta = \rho \Delta S \Delta L \frac{\Delta u}{\Delta t}$$

$$P\Delta S - P\Delta S - \Delta P \Delta S - \rho \Delta S \Delta L g \cos\beta = \rho \Delta S \Delta L \frac{\Delta u}{\Delta t}$$

Dividing through out by ρΔ S Δ L
gives

$$\frac{\Delta P}{\rho \Delta L} + g\cos\beta + \frac{\Delta u}{\Delta t} = 0$$

But from the figure $\cos\beta = \frac{\Delta z}{\Delta L}$

$$\frac{\Delta u}{\Delta t} = \frac{\Delta u}{\Delta L}\frac{\Delta L}{\Delta t} = u\frac{\Delta u}{\Delta L}$$

Substituting the value of cos β and u =ΔL/Δt, we get

$$\frac{\Delta P}{\rho \Delta L} + g\frac{\Delta z}{\Delta L} + u\frac{\Delta u}{\Delta L} = 0$$

Now taking the limit of the terms in the above equation as Δ L approaches zero we get the point form of Bernoulli's equation or Euler's equation

$$\frac{dP}{\rho dL} + g\frac{dz}{dL} + u\frac{du}{dL} = 0$$

$$\frac{1}{\rho}\frac{dP}{dL} + g\frac{dz}{dL} + \frac{d\left(\frac{1}{2}u^2\right)}{dL} = 0$$

The differential form of the equation is

$$\frac{1}{\rho}dP + gdz + d\left(\frac{1}{2}u^2\right) = 0$$

Case (i) : for incompressible fluids density is constant and the above equation can be integrated to give

$$\frac{P}{\rho} + gz + \frac{u^2}{2} = \text{constant}$$

or in the head form

$$\frac{P}{\rho g} + z + \frac{u^2}{2g} = \text{constant}$$

Between two definite points on the stream tube the above equation can be written as

$$\frac{P_1}{\rho} + gz_1 + \frac{u_1^2}{2} = \frac{P_2}{\rho} + gz_2 + \frac{u_2^2}{2}$$

$$\frac{P_1}{\rho g} + z_1 + \frac{u_1^2}{2g} = \frac{P_2}{\rho g} + z_2 + \frac{u_2^2}{2g}$$

Modification of Bernoulli's equation

For real fluids that are passing through pipe are influenced by the solid boundaries. To extend the Bernoulli's equation to cover these practical situations two modifications are required. They are

- Fluid friction correction factor
- Kinetic energy correction factor

i. Fluid friction correction factor: Fluid friction correction factor can be defined as conversion of mechanical energy into heat in the flowing stream. So, in frictional fluid the total head is not constant along a stream line and always decrease in the direction of flow. In accordance with the principle of conservation of energy an amount of heat generated is equivalent to the loss in mechanical energy. So, for incompressible fluids, the bernoulli's equation is corrected for friction by adding a term to the right hand side of the equation.

$$\frac{P_a}{\rho} + gz_a + \frac{u_a^2}{2} = \frac{P_b}{\rho} + gz_b + \frac{u_b^2}{2} + h_f$$

where h_f represents all the friction generated per unit fluid that occurs in the fluid between stations a and b.

Friction appears in boundary layers because the work done by shear forces in maintaining the velocity gradients in both laminar and turbulent flow is eventually converted into heat by viscous action. Friction generated in un separated boundary layers is called *skin friction*. When the boundary layers separate and form wakes, additional energy dissipation appears within the wake and friction of this type is called *form friction* since it is a function of the position and shape of the solid. In a given situation both skin friction and form friction may be active in varying degrees. The total friction h_f in the above equation includes both types of frictional loss.

ii. Kinetic energy correction factor : Kinetic energy correction factor is defined as the ratio of the kinetic energy of the flow per second based on actual velocity across a section to the kinetic energy of the flow per second based on average velocity across the same section. It is denoted by a. Hence mathematically

$$\alpha = \frac{\text{K.E /sec based on actual velocity}}{\text{K.E /sec based on average velocity}}$$

K.E/sec based on actual velocity across a section: Consider an element of cross-sectional area S. The mass flow rate through this is ρu ds. Each kg of fluid flowing through area ds carries kinetic energy in amount $u^2/2$ and the energy flow rate through area ds is therefore,

$$dE_k = (\rho u ds)\frac{u^2}{2} = \frac{\rho u^3}{2} ds$$

, where E_k represents the time rate of flow of kinetic energy.

total / net rate of flow of kinetic energy through the entire cross section is

$$E = \frac{\rho}{2}\int_0^s u^3 ds$$

But total rate of mass flow,

so K.E/sec based on average velocity $= \dot{m}\frac{v^2}{2}$

$$= \rho v s \frac{v^2}{2}$$

Kinetic energy correction factor

$$a = \frac{\int_0^s u^3 ds}{v^3 S}$$

Kinetic energy correction factor for laminar flow is 2 and not to be neglected

Kinetic energy correction factor for turbulent flow is ~ 1 and can be neglected

Problems with out friction

Water is flowing through a pipe of 5 cm diameter under a pressure of $2.943 * 10^5 N/m^2$(gauge) and with mean velocity of 2 m/s. Find the total head or total energy per unit mass of the water at a cross section which is 5 m above the datum line.

Data: Diameter of pipe = 5 cm = 0.05 m.

Pressure $P = 2.943*10^5\ N/m^2$

Velocity v = 2 m/s

Datum head z = 5 m

Total head = pressure head + kinetic head + datum head

pressure head= $\frac{P}{\rho g} = \frac{2.943*10^5}{1000*9.81}$ = 30m

Kinetic head = $\frac{v^2}{2g} = \frac{2^2}{2*9.81}$ =0.204m

Total head =

30+0.204+5=35.204m

A pipe through which water is flowing is having diameter 20cm and 10cm at the cross section 1 and 2 respectively. The velocity of water at section 1 is 4 m/s. Find the velocity head at section 1 and 2 and also rate of discharge.

D_1=20cm=0.20m D_2=10cm=0.10m

$S_1=(\pi/4)*D_1^2$ $S_2=(\pi/4)*D_2^2$

$=(\pi/4)*0.2^2=$ $=(\pi/4)*0.1^2=0.00785\ m^2$

$0.314m^2$

$v_1=4m/s$ $V_1=?\ m/s$

Velocity head at section 1 = $\frac{v_1^2}{2g}=\frac{4^2}{2*9.81}$ = 0.815m

We know from continuity equation for incompressible fluids that

$$S_1 v_1 = S_2 v_2$$

$$v_2 = \frac{S_1 v_1}{S_2} = \frac{0.0314 * 4}{0.00785} = 16\text{m/s}$$

Velocity head at section 2 $$\frac{v_2^2}{2g} = \frac{16^2}{2 * 9.81} = 13.047\text{m}$$

Rate of discharge = $S_1 v_1$ = 0.0314*4=**0.1256 m^3/s.**

The diameter of a pipe changes from 20cm at a section 5m above datum, to 5cm at section 3m above datum. The pressure of water at first section is $5*10^5 N/m^2$. If the velocity of flow at the first section is 1m/s, determine the intensity of pressure at the second section.

$D_1 = 20cm = 0.2m$

$S_1 = (\pi/4)*D_1^2 = (\pi/4)*0.2^2 = 0.314m^2$

$v_1 = 1m/s$

$P_1 = 5*10^5 N/m^2$

$z_{1=} 5m$

$D_2 = 5cm = 0.05m$

$S_2 = (\pi/4)*D_2^2 = (\pi/4)*0.05^2 = 1.96*^{10-3} m^2$

$v_2 = ?$

$P_2 = ?$

$z_2 = 3m$

We know from continuity equation for incompressible fluids that
$S_1v_1 = S_2v_2$

$$v_2 = \frac{S_1 v_1}{S_2} = \frac{0.0314*1}{1.96*10^{-3}}$$ =16.02 m/s

Applying Bernoulli's equation between the sections 1and 2

$$\frac{P_1}{\rho} + \frac{v_1^2}{2} + z_1 g = \frac{P_2}{\rho} + \frac{v_2^2}{2} + z_2 g$$

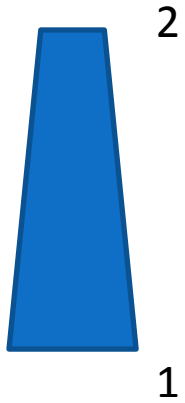
$$\frac{5*10^5}{1000} + \frac{1^2}{2} + 5*9.81 = \frac{P_2}{1000} + \frac{16^2}{2} + 3*9.81$$

549.55=157.43+(P_2/1000)
Solving for P_2 from the above equation $P_2 = 3.9212*10^5 N/m^2$.

The water is flowing through a pipe having diameters 20cm and 10cm at section 1 and 2 respectively. The rate of flow through pipe is 35 lit/s. the section 1 is 6m above datum and section 2 is 4m above datum. If the pressure at section 1 is $3.924*10^5 N/m^2$. Find the intensity of pressure at section 2.

D_1=20cm = 0.2m

D_2=10cm = 0.1m



$S_1=(\pi/4)*D_1^2$ = $(\pi/4)*0.2^2$ = $0.0314 m^2$

$S_2=(\pi/4)*D_2^2$ = $(\pi/4)*0.1^2$= 0.00782 m^2

Q = 35lit/s=0.035 m^3/s

v_2 = ?

$P_1=3.924*10^5 N/m^2$

P_2 = ?

z_1= 6m

z_2 = 4m

$$v_1 = \frac{Q}{S_1} = \frac{0.035}{0.0314} = 1.114 \text{ m/s}$$

$$v_2 = \frac{Q}{S_2} = \frac{0.035}{0.00785} = 4.456 \text{ m/s}$$

Applying Bernoulli's equation between sections 1 and 2

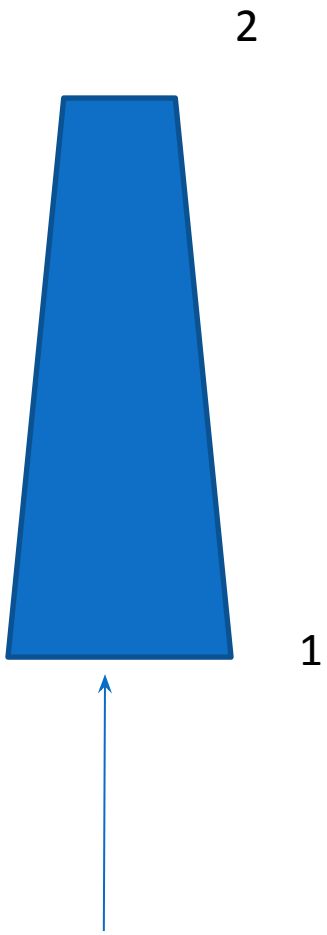
$$\frac{P_1}{\rho} + \frac{v_1^2}{2} + z_1 g = \frac{P_2}{\rho} + \frac{v_2^2}{2} + z_2 g$$

$$\frac{3.924*10^5}{1000} + \frac{1.114^2}{2} + 6*9.81 = \frac{P_2}{1000} + \frac{4.456^2}{2} + 4*9.81$$

$$P_2 = 4.027*10^5$$

Water is flowing through a pipe having diameter 30cm and 20cm at the bottom and upper end respectively. The intensity of pressure at the bottom end is $2.4525*10^5$ N/m^2 and the pressure at the upper end is $0.981*10^5$ N/m^2. Determine the difference in datum{level} if the rate of flow through pipe is 40lit/s.

$Q = 0.04m^3/s = S_1v_1 = S_2v_2$



D_1=30cm = 0.3m

D_2=20cm = 0.2m

$S_1=(\pi/4)*D_1^2$ =

$(\pi/4)*0.3^2 = 0.0707m^2$

$S_2=(\pi/4)*D_2^2$ =

$(\pi/4)*0.2^2= 0.0314\ m^2$

Q = 40lit/s=0.040 m^3/s

v_2 = ?

$P_1=2.4525*10^5 N/m^2$

$P_2 = 0.981*10^5$ N/m^2

$$v_1 = \frac{Q}{S_1} = \frac{0.04}{0.0707} = 0.5658 \text{ m/s}$$

$$v_2 = \frac{Q}{S_2} = \frac{0.04}{0.0314} = 1.274 \text{ m/s}$$

Applying Bernoulli's equation in terms of head

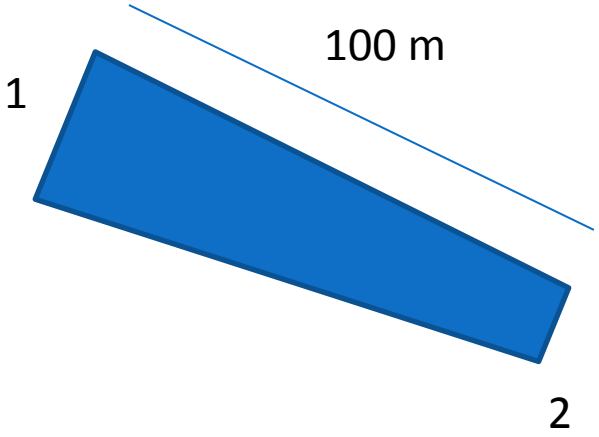
$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

$$\frac{2.4525 * 10^5}{1000 * 9.81} + \frac{0.5658^2}{2 * 9.81} + z_1 = \frac{0.981 * 10^5}{1000 * 9.81} + \frac{1.274^2}{2 * 9.81} + z_2$$

Solving for $\mathbf{z_2 - z_1 = 14.93m}$

Water is flowing through a tapering pipe of length 100m having diameter 60cm at the upper end and 30cm at the lower end, at the rate of 50lit/s. The pipe has a slope of 1 in 30. Find the pressure at the lower end, if the pressure at the higher level is $1.962*10^5 N/m^2$.

Data : Slope of the pipe (axis) is 1 in 30.



D_1=60cm = 0.6m

D_2=30cm = 0.3m

$S_1=(\pi/4)*D_1^2$ = $(\pi/4)*0.6^2$ = $0.2827 m^2$

$S_2=(\pi/4)*D_2^2$ = $(\pi/4)*0.3^2$= 0.07068 m^2

Q = 50lit/s=0.050 m^3/s

v_2 = ?

$P_1=1.962*10^5 N/m^2$

P_2 = ? N/m^2

Let the datum line pass through the center line of the lower end. i.e., $z_2=0$.
For every 30m length of pipe – height decreases by 1m
Hence for 100m of pipe decrease in height

$$\left(\frac{1}{30}*100\right)$$

$$=3.33m$$

$Q = 0.05m^3/s = S_1v_1 = S_2v_2$

$$v_1 = \frac{Q}{S_1} = \frac{0.05}{0.2827} \quad = 0.1768 \text{ m/s}$$

$$v_2 = \frac{Q}{S_2} = \frac{0.05}{0.07068} \quad = 0.7074 \text{ m/s}$$

Applying Bernoulli's equation between upper and lower sections of pipe

$$\frac{P_1}{\rho} + \frac{v_1^2}{2} + z_1 g = \frac{P_2}{\rho} + \frac{v_2^2}{2} + z_2 g$$

$$\frac{1.962 * 10^5}{1000} + \frac{0.1768^2}{2} + 3.333 * 9.81 = \frac{P_2}{1000} + \frac{0.07068^2}{2} + 0$$

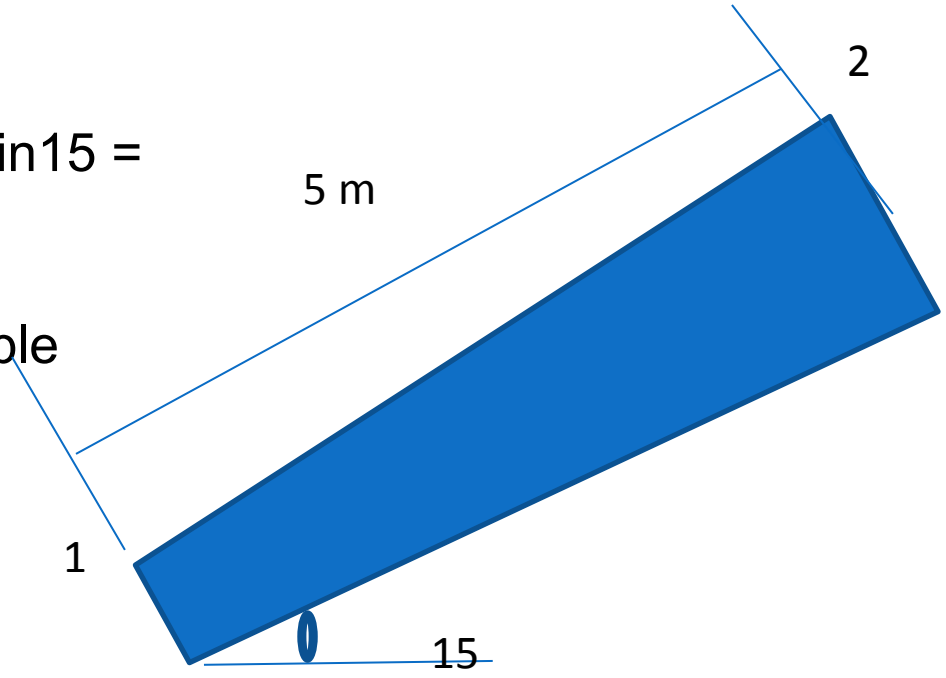
Solving for $P_2 = 2.28909 * 10^5 N/m^2$

A 5m long pipe in inclined at an angle of 15° with the horizontal. The smaller section of the pipe, which is at a lower level is of 80mm dia and the larger section of the pipe is of 240mm dia as shown in figure. Determine the difference of pressure between the two sections in N/m^2 if the pipe is uniformly tapered and the velocity of water at the smaller section is 1m/s.

z_1=0m; z_2=0+5sinϴ = 5 sin15 = 1.294m.

We know from continuity equation for incompressible fluids that

$S_1v_1 = S_2v_2$



D_1=80mm = 0.08m D_2=240mm = 0.24m

$S_1=(\pi/4)*D_1^2$ = $S_2=(\pi/4)*D_2^2$ =

$(\pi/4)*0.08^2$ = $(\pi/4)*0.24^2$= 0.042

$0.005027m^2$ m^2

v_1= 1m/s v_2 = ?

z_1=0m; z_2=0+5sina = 5 sin15 = 1.294m.

We know from continuity equation for incompressible fluids that

$S_1v_1 = S_2v_2$

$$v_2 = \frac{S_1 v_1}{S_2} = \frac{0.005027 * 1}{0.0452}$$

=0.1112 m/s

Applying Bernoulli's equation between the sections 1and 2

$$\frac{P_1}{\rho} + \frac{v_1^2}{2} + z_1 g = \frac{P_2}{\rho} + \frac{v_2^2}{2} + z_2 g$$

$$\frac{P_1 - P_2}{\rho} = \frac{v_2^2 - v_1^2}{2} + g(z_2 - z_1)$$

$$\frac{P_1 - P_2}{\rho} = \frac{0.1112^2 - 1^2}{2} + 9.81(1.294 - 0)$$

$$\frac{P_1 - P_2}{\rho} = 12.2003$$

P_1-P_2 = 12200.32N/m^2 = 12.20032kN/ m^2

A pipe of diameter 40cm carriers water at a velocity of 25m/s.The pressure, at the points A and B are given as $2.943*10^5 N/m^2$ and $2.254*10^5 N/m^2$ respectively, while the datum head at A and B are 28m and 30m. Find the loss of head between A and B.

Data: Diameter of pipe, d= 40cm =0.4m

Velocity, v = 25m/s

$P_A = 2.943*10^5\ N/m^2$ $P_B = 2.254*10^5 N/m^2$

$z_A = 28m$ $z_B = 30m$

Applying Bernoulli's equation between sections A and B in head form

$$\frac{P_A}{\rho g} + \frac{v_A^2}{2g} + z_A = \frac{P_B}{\rho g} + \frac{v_B^2}{2g} + z_B + h_f$$

$$h_f = \left(\frac{P_A}{\rho g} + \frac{v_A^2}{2g} + z_A \right) - \left(\frac{P_B}{\rho g} + \frac{v_B^2}{2g} + z_B \right)$$

$$h_f = \left(\frac{2.943 * 10^5}{1000 * 9.81} + \frac{25^2}{2 * 9.81} + 28 \right) - \left(\frac{2.254 * 10^5}{1000 * 9.81} + \frac{25^2}{2 * 9.81} + 30 \right)$$

h_f = 5.023m of water. (loss in head)

A conical tube of length 2m is fixed vertically with its smaller end upwards. The velocity of flow at the smaller end is 5m/s while at the lower end it is 2m/s. The pressure head at the smaller end is 2.5m of liquid. The loss of head in the tube is

$$\frac{0.35(v_1 - v_2)^2}{2g} = h_L$$

where v_1 is the velocity at the smaller end and v_2 at the lower end respectively. Determine the pressure head at the lower end. Flow takes place in the downward direction.

Data: Let upper end be 1 and lower end be 2; L=2m

$\frac{P_1}{\rho g}$ = 2.5m v_1= 5m/s; v_2= 2m/s;

$$h_L = \frac{0.35(v_1 - v_2)^2}{2g} = \frac{0.35*(5-2)^2}{2*9.81}$$

loss of
head=0.16m

Let the datum line pass through the section 2, Applying Bernoulli's equation

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

$$2.5 + \frac{5^2}{2*9.81} + 2 = \frac{P_2}{\rho g} + \frac{2^2}{2*9.81} + 0 + 0.16$$

$$5.7742 = \frac{P_2}{\rho g} + 0.3639$$

$$\frac{P_2}{\rho g} = 5.4103\text{m of liquid.}$$

A pipe line carrying oil of specific gravity 0.87 changes in diameter from 200mm diameter at a position A to 500mm at a position B which is 4m at a higher level. If the pressure at A and B are $0.981*10^5 N/m^2$ and $0.5886*10^5 N/m^2$ respectively, and the discharge is 200lit/s, determine the loss of head and direction of flow.

$D_A = 200mm = 0.2m$ $D_B = 500mm = 0.5m$

$S_A=(\pi/4)*D_A^2 = (\pi/4)*0.2^2 = 0.0314m^2$ $S_B=(\pi/4)*D_B^2 = (\pi/4)*0.5^2= 0.1963\ m^2$

$P_A = 0.981*10^5\ N/m^2$ $P_B = 0.5886*10^5\ N/m^2$

$Q = 200\ lit/s = 0.2m^3/s$

Let the datum line pass through A
$z_A = 0m$ and $z_B = 4m$

we know $$v_A = \frac{Q}{S_A} = \frac{0.2}{0.0314} = 6.369 m/s$$

$$v_B = \frac{Q}{S_B} = \frac{0.2}{0.1963} = 1.018 m/s$$

Applying Bernoulli's equation between A and B

$$\frac{P_A}{\rho} + \frac{v_A^2}{2} + gz_A = \frac{P_B}{\rho} + \frac{v_B^2}{2} + gz_B + h_f$$

$$\frac{0.981*10^5}{870}+\frac{6.369^2}{2}+9.81*0=\frac{0.5886*10^5}{870}+\frac{1.018^2}{2}+9.81*4+h_f$$

Solving for h_f,

h_f = 25.63J/kg.

The +ve sign signifies that the total energy at section Ais greater than at section B. the flow takes place from bottom to top to (A to B).

The loss of energy h_f = **25.63J/kg** =25.63/9.81m = **2.613m** of liquid.

Pump problems:

Work supplied to the pump from shaft work = w_s

Pump work = w_p

Total friction in the pump/kg of fluid = h_{fp}

Net mechanical energy available to the flowing fluid = $w_p - h_{fp}$

Pump efficiency η = $(w_p - h_{fp}) / w_s$

The mechanical energy delivered to the flowing fluid = $\eta w \dot{m}$

Power, P = $\eta w_p \dot{m}$ Watts

or

$$\frac{\eta \omega_p m}{746} hp$$

Applying the Bernoulli's equation around the pump, the pressure drop developed by the pump is calculated as follows

$$\frac{P_1}{\rho} + gz_1 + \frac{V_1^2}{2} + \eta w_p = \frac{P_2}{\rho} + gz_2 + \frac{V_2^2}{2}$$

here $z_1 = z_2$ (as pump is parallel to datum line)

$V_1 = V_2$ (rate of flow is steady and dia of the pipe on suction side and delivery side are same)

$$\frac{P_2}{\rho} - \frac{P_1}{\rho} = \eta w_p$$

if $V_1 \neq V_2$

$$\frac{P_2}{\rho} - \frac{P_1}{\rho} = \eta w_p + \left(\frac{V_1^2 - V_2^2}{2} \right)$$

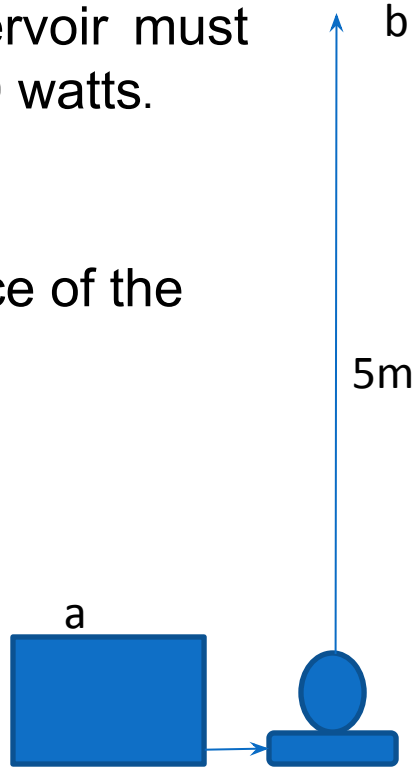
Water at 25°C is pumped at a constant rate of $24*10^{-4}m^3/s$ from a large reservoir resting on the floor to the top of an experimental absorption tower. The point of discharge is 5m above the floor and the frictional loss in the 0.05m pipe from the reservoir to the tower amounts to 2.4J/kg. At what height in the reservoir must the water level be kept if the pump can develop only 100 watts.

$\rho_{water} = 1000\ kg/m^3$

Floor is taken as datum line. Let station 'a' be the surface of the water in the reservoir and 'b' the discharge point.

Applying Bernoulli's equation between station a and b

$$\frac{P_a}{\rho} + \frac{v_a^2}{2} + gz_a + \eta W_p = \frac{P_b}{\rho} + \frac{v_b^2}{2} + gz_b + h_f$$



The velocity v_a in the reservoir will be very small and can be neglected. The pressure P_a and P_b is atmospheric. So they get cancelled out.

We know Power, $P = \dot{m}\, W_p$ 100W $\Rightarrow$ $W_p = 100 / \dot{m}$

Where $\dot{m} = Q * \rho = 24 * 10^{-4} * 1000 = 2.4 kg/s$ $\Leftarrow$

$$w_p = \frac{100}{2.4} = 41.67 KJ/kg$$

Area of the delivery pipe $S_b = (pie/4)*d^2 = (pie/4)*0.05^2 = 1.96*10^{-3} m^2$

$v_b = Q/S_b = (24*10^{-4})/(1.96*10^{-3}) = 1.22 m/s$

$z_b = 5m$ and $h_f = 2.4 J/kg$

Substituting above values in Bernoulli's equation

$$z_a + 41.67/9.8 = \frac{1.22^2}{2*9.81} + 5 + 2.4/9.8$$

$\Longrightarrow z_a > 1.06$

A height of at least 1.06 m should be maintained in the feed tank for the given operation.

Oil of specific gravity 0.75 is pumped from a tank over a hill through a 60cm pipe with the pressure at the top of the hill maintained at $1.75*10^5 N/m^2$. The summit of the hill is 80m above the surface of the oil in the tank and oil is pumped at the rate of $0.6m^3/s$. If the lost head from the tank to summit is 5m, what HP must the pump supply to the liquid?

Data :$P_b = 1.75*10^5 N/m^2$.

$P_a = 1.01325*10^5 N/m^2$ (atmospheric)

$h_f = 5m = 5*9.81 = 49.05 J/kg$

Let the station 'a' be the surface of the oil in the reservoir and station 'b' be the discharge point. Let the datum line pass through the surface of the oil.

$z_a = 0$; $z_b = 80m$

Discharge, $Q=0.6m^3/s$; diameter of pipe =0.6m

Area of pipe, $S_b = (\pi/4)*0.6^2 = 0.283m^2$.

$v_b = Q/S_b = 0.6/0.283 = 2.12m/s$

v_a is very small, because of the larger cross-section of the reservoir

$\frac{v_a^2}{2g}$ can be neglected.

Applying Bernoulli's equation

$$\frac{P_a}{\rho} + \frac{v_a^2}{2} + gz_a + \eta W_p = \frac{P_b}{\rho} + \frac{v_b^2}{2} + gz_b + h_f$$

$$\frac{1.0132\mathrm{E}*10^5}{750} + 9.81*0 + \eta W_p = \frac{1.75*10^5}{750} + \frac{2.12^2}{2} + 9.81*80 + 49.05$$

$\eta W_p = 935.33$ J/kg

Power, $P = \dot{m}\,\eta W_p$

$\dot{m} = Q\rho = 0.6*750 = 450$ kg/s

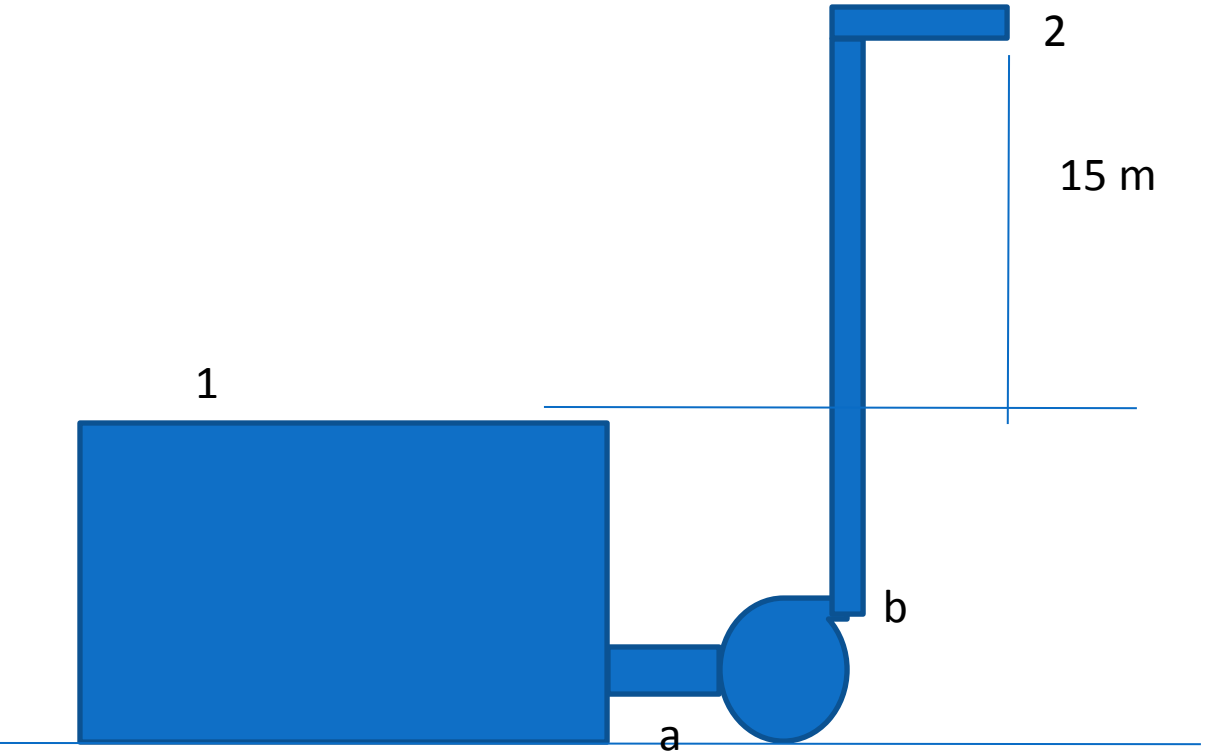
P = 450 * 935.33 = 420453.5 W

P = 314498.5/746 = **563.26 HP**

A Pump is used to draw water from a storage tank through a 100 mm pipe. The efficiency of pump is 60% the velocity in the suction line is 1 m/s. Pump discharges through a 50mm pipe to an overhead tank. The end of the discharge line is 15m above the level of the water. If the loss in the entire piping system is 30J/Kg what is the power required for pumping. what pressure must the pump develop

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + \eta W_p = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

Pa– Pb=?



D_1=100mm = .1m

D_2=50mm = 0.05m

$S_1=(\pi/4)*D_1^2 = (\pi/4)*.1^2$

$S_2=(\pi/4)*0.05^2$

V_a=1 m/sec V_1=0

$v_b = v_2$ = 4m/sec

P_1=1 atm

P_2 = 1 atm

Pa– Pb=?

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + \eta W_p = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

$$W_p = \frac{P_2}{\rho g} - \frac{P_1}{\rho g} + \frac{v_2^2}{2g} - \frac{v_1^2}{2g}$$

$V_1 =$ P1=P2

0

$$\eta W_p = \frac{v_2^2}{2g} + z_2 - z_1 + h_L$$

$$\eta W_p = \frac{4^2}{2*9.8} + 15 + \frac{30}{9.8}$$

$$\eta W_p = 18.87$$

$$\eta = 0.6$$

$$W_p = 31.45m$$

$$Power = \dot{m} g W_p$$

$$\dot{m} = \rho V S = 7.85$$

$$Power = 7.85 * 9.81 * 31.45$$

2421.9 1735watts

=3.2465 hp

Appyling Bernoullis equation between points a (inlet)and b (Exit)

ie between the inlet and outlet of the pump

$$\frac{P_a}{\rho g} + \frac{v_a^2}{2g} + z_a + \eta W_p = \frac{Pb}{\rho g} + \frac{v_b^2}{2g} + z_b$$

Za=Zb

$$\frac{P_a}{\rho g} - \frac{Pb}{\rho g} = \frac{16}{2g} - \frac{1}{2g} - 31.45$$

=.765-31.45= -30.68 Exit pressure is more

m

$P_b - P_a = 30.68 \times 9.8 \times 1000$

$= 3 * 10^5$ N/m^2